JUNE · IAS 2026 · CLASS 12

The Interview, Part One

Introduction and Methods — in your own words, as if someone asked you after the talk.

PART 1

Introduction

Why this question matters, and what nobody knew yet.

Q1

Let's start simple. What is chronic ankle instability, and how common is it after a first ankle sprain?

Aim for about a minute.

INTRODUCTION · Q1 · ONE WAY TO SAY IT

Let's start simple. What is chronic ankle instability, and how common is it after a first ankle sprain?

Chronic ankle instability, or CAI, is a condition that develops after a lateral ankle sprain. People with CAI have **repeated episodes of giving way**, and they often feel the ankle is unstable. It's very common — **up to 40% of** people who have a first lateral ankle sprain **go on to develop** CAI.

KEY EXPRESSIONS

repeated episodes of ~ · **up to ~% of ~** · **go on to develop ~**

Q2

What kinds of problems do people with CAI have with balance — and why are dynamic tasks like landing especially important?

Aim for about a minute.

What kinds of problems do people with CAI have with balance — and why are dynamic tasks like landing especially important?

One of the main features of CAI is impaired balance. It**'s linked to** deficits in postural control after the ligament injury. And the problem **is most visible in** dynamic tasks, like jump landing. **That matters because** poor control during landing may increase the risk of re-injury.

KEY EXPRESSIONS

One of the main features of ~ is ~ · **is linked to ~** · **is most visible in ~** · **That matters because ~**

Q3

Your study is about the ankle, but you're looking at the "core." Why look beyond the ankle itself?

Aim for about a minute.

INTRODUCTION · Q3 · ONE WAY TO SAY IT

Your study is about the ankle, but you're looking at the "core." Why look beyond the ankle itself?

Because postural control **depends on more than** the ankle. It also depends on sensorimotor control and core stability. The core provides proximal stability for distal mobility — **in other words**, a stable center lets the limbs move well. And this idea **can be extended to** the foot. **So we looked at** both the trunk core and the foot core.

KEY EXPRESSIONS

depends on more than ~ · **in other words** · **can be extended to ~** · **So we looked at ~**

Q4

You mention two models — Panjabi's trunk core and McKeon's foot core. Can you explain each one, and what they have in common?

Aim for about a minute.

INTRODUCTION · Q4 · ONE WAY TO SAY IT

You mention two models — Panjabi's trunk core and McKeon's foot core. Can you explain each one, and what they have in common?

Panjabi **describes** the trunk core **as** three subsystems — passive, active, and neural — that work together to keep the spine stable. McKeon **applied the same idea to** the foot. The foot core also has passive, active, and neural subsystems. **What they have in common is that** if one subsystem fails, overall stability is compromised. Together, they **give us a framework for** looking at both the proximal and the distal core.

KEY EXPRESSIONS

describes A as B · **applied the same idea to ~** · **What they have in common is that ~** · **give us a framework for ~**

Q5

What does the previous research say about deficits at the foot level and at the trunk level in people with CAI?

Aim for about a minute.

INTRODUCTION · Q5 · ONE WAY TO SAY IT

What does the previous research say about deficits at the foot level and at the trunk level in people with CAI?

At the foot level, people with CAI show weaker intrinsic foot muscles — atrophy and reduced toe flexor strength — and that **has been linked to** poor balance. **At the trunk level**, studies **have reported** reduced contractility of deep trunk muscles, like the transversus abdominis and the diaphragm. So there are deficits at both ends, and core weakness is considered **a major contributing factor to** CAI.

KEY EXPRESSIONS

At the ~ level · **has been linked to ~** · **Studies have reported ~** · **a major contributing factor to ~**

Q6

Both types of training have been studied before. So what's the gap — what did nobody know until your study, and what exactly did you set out to compare?

Aim for about a minute.

INTRODUCTION · Q6 · ONE WAY TO SAY IT

Both types of training have been studied before. So what's the gap — what did nobody know until your study, and what exactly did you set out to compare?

Both types of training **have been shown to** improve postural control in CAI. But two things were unclear. First, whether those effects **carry over to** more dynamic, sport-like tasks such as landing. Second, the two approaches come from different ideas — **bottom-up versus top-down** — but very few studies **have compared them directly**. **So we compared** foot core training, trunk core training, and a control group on static balance, dynamic balance, and lower-extremity and trunk kinematics during landing.

KEY EXPRESSIONS

have been shown to ~ · **carry over to ~** · **A versus B** · **compare them directly** · **So we compared ~**

PART 2

Methods

What you did — clearly and briefly.

Q1

Walk me through the overall design of the study in one or two sentences.

Aim for 30 to 60 seconds.

METHODS · Q1 · ONE WAY TO SAY IT

Walk me through the overall design of the study in one or two sentences.

It was a **randomized, three-group, pretest–posttest design**. We assessed everyone **before and after** a four-week intervention.

KEY EXPRESSIONS

a randomized, three-group, pretest–posttest design · **before and after ~**

Q2

Who could take part? How did you decide someone had CAI, and who did you exclude?

Aim for 30 to 60 seconds.

METHODS · Q2 · ONE WAY TO SAY IT

Who could take part? How did you decide someone had CAI, and who did you exclude?

We followed the International Ankle Consortium **criteria**. Participants needed **at least** one significant lateral ankle sprain, with the first one at least a year ago, at least two episodes of giving way in the past six months, and IdFAI and FAAM scores that showed functional instability. **We excluded anyone with** previous lower-limb surgery, an acute injury in the past three months, or a supervised exercise program in the past month.

KEY EXPRESSIONS

We followed ~ criteria · **at least ~** · **We excluded anyone with ~**

Q3

How did you arrive at thirty participants, and how did you assign them to the three groups?

Aim for 30 to 60 seconds.

METHODS · Q3 · ONE WAY TO SAY IT

How did you arrive at thirty participants, and how did you assign them to the three groups?

We ran **an a priori power analysis** in G*Power, which showed we needed thirty participants. We **stratified** them **by sex** and **randomly assigned** them **to** three groups of ten — foot core, trunk core, and control — using a computer-generated sequence.

KEY EXPRESSIONS

an a priori power analysis · **stratified by ~** · **randomly assigned to ~**

Q4

How did you measure static balance — what did participants actually do, and what did you calculate from it?

Aim for 30 to 60 seconds.

METHODS · Q4 · ONE WAY TO SAY IT

How did you measure static balance — what did participants actually do, and what did you calculate from it?

Participants **stood barefoot on the involved leg**, hands on the waist, for ten seconds — **with eyes open and eyes closed**, three trials each. **From** the force plate, **we calculated** center-of-pressure mean velocity, in total and in each direction, and the 95% ellipse area.

KEY EXPRESSIONS

stood barefoot on the involved leg · **with eyes open and eyes closed** · **From ~, we calculated ~**

Q5

Now the landing task. Can you describe it step by step, and explain what the DPSI tells you?

Aim for 30 to 60 seconds.

METHODS · Q5 · ONE WAY TO SAY IT

Now the landing task. Can you describe it step by step, and explain what the DPSI tells you?

Participants started at 40% of their height from the force plate, **jumped forward over** a 30-centimeter hurdle, and **landed on** the involved leg, holding the position for ten seconds. The DPSI **combines** ground reaction forces **in three directions** over the first three seconds after landing. **A lower score means** better stability.

KEY EXPRESSIONS

jumped forward over ~ and landed on ~ · **combines ~ in three directions** · **A lower score means ~**

Q6

How did you capture the movement during landing? How did you define initial contact, and why did you focus on the first 200 milliseconds?

Aim for 30 to 60 seconds.

METHODS · Q6 · ONE WAY TO SAY IT

How did you capture the movement during landing? How did you define initial contact, and why did you focus on the first 200 milliseconds?

We used a Vicon motion-capture system with 35 reflective markers, **synchronized with** the force plate. Initial contact **was defined as** the first frame where vertical force went above 20 newtons. **We focused on** the first 200 milliseconds **because** that's when the body has to absorb the landing and control the joints.

KEY EXPRESSIONS

synchronized with ~ · **was defined as ~** · **We focused on ~ because ~**

※ 200ms 를 고른 이유는 원고에 없어서 일반적인 근거로 넣었어요 — June 에게 실제 이유 확인

Q7

What did the two training groups actually do? Give me a couple of example exercises from each program, and tell me how often they trained.

Aim for 30 to 60 seconds.

METHODS · Q7 · ONE WAY TO SAY IT

What did the two training groups actually do? Give me a couple of example exercises from each program, and tell me how often they trained.

Both groups trained **three times a week for four weeks** — twelve supervised sessions of about thirty minutes. The foot core group did exercises that **target** the intrinsic foot muscles, like the short-foot exercise and towel curls. The trunk core group did exercises for the deep abdominal muscles, like the abdominal drawing-in maneuver and planks. Both programs **got progressively harder** over the four weeks.

KEY EXPRESSIONS

three times a week for four weeks · **target ~** · **got progressively harder**

Q8

Briefly, how did you analyze the data? Why did you use SPM in addition to the standard ANOVA?

Aim for 30 to 60 seconds.

METHODS · Q8 · ONE WAY TO SAY IT

Briefly, how did you analyze the data? Why did you use SPM in addition to the standard ANOVA?

We used a two-way mixed ANOVA **to look at** time, group, and their interaction. For the kinematic data, we also used statistical parametric mapping, or SPM. **The reason is that** ANOVA **only looks at** single points, like the angle at initial contact, **while** SPM looks at the whole 200-millisecond curve, **so we don't miss** changes that happen between those points.

KEY EXPRESSIONS

to look at ~ · **The reason is that ~** · **only looks at A, while B ~** · **so we don't miss ~**